

One Operator Is Not Enough — Yet: The Structural Cost of a Single-Operator Mathematical Representation, Measured at Fixed Scale with Published Nulls

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Abstract

Odrzywołek (Odrzywołek, 2026) showed that every elementary function is constructible from the constant 1 and a single binary operator, $\text{eml}(x, y) = \exp(x) - \ln(y)$, making any expression a strict binary tree whose internal nodes all carry the same operation. A single-operator representation is maximally regular, which is attractive for graph learning; the obvious cost is size. Before asking whether the regularity helps a model, we measure the cost exactly, at one fixed scale: a deterministic, quality-audited corpus of 250,000 expressions with frozen splits. Compiling the corpus with the pinned official compiler yields a median tree expansion factor $\alpha = 40.66$ over the source AST (mean 952.1, p99 10,448.6); 0/250,000 expressions pass the preregistered per-family thresholds (1.29–1.50; the whole-grammar counting break-even is ≈ 1.56). Lossless DAG sharing compresses pure-EML trees $39.4\times$ on average, yet the shared EML DAG is larger than the *uncompressed* AST tree for every one of the 250,000 expressions (best remaining ratio exactly $8/7$). Bounded, verified equality saturation improves 23.9–27.6% of costed expressions at 2.4–2.8% mean relative savings, conditional on 60.7% vocabulary coverage. We publish two preregistered nulls — a learned motif vocabulary loses to an equal-budget frequent baseline, and a neural cost ranker loses to a structural heuristic — and one preregistered gate verdict, a *fail*, reported with content hashes rather than softened. We also release a frozen, leakage-controlled learning apparatus (equivalence, rewrite-step, proof/simplification, and symbolic-regression tracks) whose training verdicts are the paper’s only pending numbers; the pair pipeline has already run end-to-end on a byte-identical pilot corpus. The study estimates no scaling laws, and frontier LLMs

appear only as external context.

1 Introduction

A single-operator representation of mathematics is maximally regular: a graph network reading a pure-EML term never has to learn *what* an operator is, only *where* things connect. The obvious objection is size — lowering even a small trigonometric expression such as $\sin(x)$ to pure EML produces hundreds of nodes — and the standard reply is that sharing (DAGs), macros, and semantic rewriting could close the gap. Whether they actually do is an empirical question that, to our knowledge, had not been measured exactly at a nontrivial scale. The compression program exists to close that gap; the learning program asks whether whatever gap remains is worth paying for as inductive bias.

This paper is deliberately narrow. It is a *structural feasibility study* under preregistered thresholds and complete denominators, publishing null and failed outcomes alongside positives, and it separates the two questions cleanly: what the representation *costs* structurally (which we measure, in full, at production scale) from whether that cost *buys* anything for learning (which we do not yet answer, and do not pretend to). We fit no scaling law, run no leaderboard, and use no frontier LLM as a controlled baseline. Every empirical sentence carries a claim-register identifier, and every number below matches a committed per-goal summary exactly.

Contributions, each tied to a claim-register entry: **C1**, a deterministic quality-audited 250,000-expression corpus with frozen splits and checksummed manifests (§2); **C2**, the exact expansion result — median $\alpha = 40.66$, 0/250,000 preregistered threshold passes (§6); **C3**, exact lossless DAG sharing — $39.4\times$ mean

EML self-compression, AST parity on zero expressions (§6); **C4**, verified e-graph rewriting (Willsey et al., 2021) and motif compression under honest denominators, including two pre-registered nulls (§6); **C9**, a bounded compiler-conformance audit whose preregistered gate verdict is a published **fail**, with artifact content hashes (§5.1); and a frozen, CI-tested, leakage-controlled learning apparatus (§§3–4), whose training verdicts (C5–C8, C10–C11) are the paper’s only pending numbers and are reported as `insufficient_evidence` until the runs land.

2 Representations, immutable data, and hypotheses

2.1 Representations

We compare a fixed family of representations, all derived from the same source expression: the ordered source AST as a tree and as an exact hash-consed DAG; the official-v4 pure-EML tree and DAG; transparent macro and motif-aware graphs (macro-derived compression vocabularies, never described as single-operator EML); a prefix-sequence control; and a transparent feature-count floor. Every graph preserves roots, ordered child slots, edge directions and roles, repeated references, exact values, and representation identity. Structural identity is kept strictly separate from semantic equivalence: two expressions share structure only when their canonical ordered signatures match, and equivalence is a distinct, verifier-established relation used only where we say so.

2.2 Fixed data foundation (C1)

The corpus contains 250,000 unique expressions with splits of exactly 175,000 / 25,000 / 25,000 / 25,000 (train / validation / test_iid / test_ood) and six source families with exact quotas: algebraic core (70,000), exp/log (40,000), powers/division/rationals (40,000), trig/hyperbolic (40,000), mixed elementary (35,000), and an OOD-stress family (25,000).¹ Generation is deterministic and QA-gated; manifests carry per-shard checksums, the config hash, seed, and clean commit. The `ood_stress` view is a

¹docs/goals/goal2/GOAL2_SUMMARY.md,
“Corpus and denominators”; FINAL_REPORT.md
#c1-goals-1-5-corpus (pre-authentication
digest index).

stored combined stress profile over size, depth, variable count, and composition, and is never relabeled strict depth-OD. Corpus v2/v3 and any regeneration of the Goals 1–5 inputs are deliberately deferred: this study is fixed to the immutable v1 foundation.

2.3 Predeclared outcome families

Outcome families were declared before the production runs and are never collapsed into one scalar leaderboard. They are: *structural* (tree and DAG size and depth, pure-EML α where compatible, macro size, dictionary-inclusive motif MDL); *predictive* (six-arm equivalence metrics); *rewrite/proof* (action and exact-successor match, verifier-valid safety, exact-target proof, simplification cost); *SR* (exact and verifier-confirmed recovery); and *compiler-v2 conformance* only. Per-family expansion thresholds were fixed in advance. Structural statistics below use all 250,000 rows; semantic auditing is a separate bounded sample (§7).

3 Compact models and fair compute

This section and §4 describe committed, CI-tested machinery whose production training has not been run; no number in them is a result. We describe the apparatus because it is frozen and because its design — not its yet-unrun output — is a contribution.

3.1 A compact GINE-style graph encoder

The four graph arms share one project-specific, compact edge-aware GINE-style encoder (Xu et al., 2019; Hu et al., 2020) — not an exact reproduction of any prior architecture. It has three edge-aware message-passing layers; hidden width 64 or 96, frozen once by measured parameter/FLOP matching; roughly 0.2–1.0 million total task parameters; node-kind, node-label, and exact-value category/encoding embeddings; edge-direction, edge-role, and ordered-slot embeddings; residual connections, GraphNorm or LayerNorm, dropout, a compact two-layer FFN, and sum pooling; and an optional virtual node whose setting is frozen before any test evaluation. For a message edge $u \rightarrow v$ the edge feature and update are

$$a_{uv} = E_{\text{dir}}(d_{uv}) + E_{\text{slot}}(s_{uv}) + E_{\text{role}}(r_{uv}), \quad (1)$$

$$\begin{aligned}\tilde{h}_v^{(\ell+1)} &= \text{MLP}_\ell \left((1 + \epsilon_\ell) h_v^{(\ell)} + m_v^{(\ell)} \right), \\ m_v^{(\ell)} &= \sum_{u \in \mathcal{N}(v)} \text{ReLU} \left(h_u^{(\ell)} + W_\ell a_{uv} \right),\end{aligned}\quad (2)$$

with the residual/normalization/dropout/FFN ordering fixed by the frozen implementation. The four arms differ only by input channel: source AST-DAG, pure-EML-DAG, an honestly labeled macro-derived frequent-motif DAG, and a train-only motif-AST fair control capped by both the reference vocabulary entry limit and exact dictionary MDL bits, with every compressed graph required to reconstruct.

3.2 Task composition and controls

For symmetric equivalence classification, shared endpoint encoders produce graph embeddings g_a, g_b combined as

$$p(a, b) = [g_a + g_b, |g_a - g_b|, g_a \odot g_b], \quad (3)$$

which is symmetric in a, b by construction; we report the swap-invariance test rather than assuming it. Rewrite-policy, value, and proof heads are directional and instead use goal-conditioned, order-sensitive features. The prefix transformer is a real matched control over a prefix token sequence, not a straw man. The transparent floor uses exactly 19 predeclared surface counts — operator, primitive, variable, size, and depth counts — that every learned arm must beat; we state explicitly that representation-specific vocabulary embeddings count toward the matched parameter budget, because pure-EML label vocabularies differ per channel and move the totals.

3.3 Compute matching and training

Because pure-EML graphs are much larger than AST or macro graphs for the same expression, equal parameters are *not* equal compute; we therefore report, jointly with every quality number, the full compute envelope: parameters, measured and estimated FLOPs under labeled methods, wall time, peak host and device memory, GPU-hours, graph/token size, effective examples and nodes per update, node/edge batch budget, gradient accumulation, the optimizer-step and example budget, optimizer, early-stopping state, precision, and deterministic settings. Hidden width and the virtual-node

Width	Virt. node	Params	Target
64	no	185,732	below
64	yes	210,692	within
96	no	343,044	within
96	yes	398,916	within

Table 1: Measured encoder parameter counts on the fixture vocabulary (Goal 6 preflight). Width-64 without a virtual node sits marginally under the $\approx 0.2\text{M}$ lower bound; this is recorded, not padded to a round number. Production vocabularies will move these totals, which is exactly why vocabulary embeddings count toward the match. The final freeze is one measured choice, not a tuned one.

flag are frozen only by a measured parameter/FLOP preflight on production vocabularies, which do not exist yet, so the grid config records `pending_preflight_freeze` rather than a guess. Table 1 gives the current fixture-vocabulary measurements that bound the freeze. We use seeds 20260726, 20260727, and 20260728 and publish every seed row; the mean never replaces the raw rows. The default production profile is $2 \times \text{H100}$ 80 GB, with $4 \times \text{H100}$ used only after a measured throughput pilot justifies it.

4 Bounded tasks and verification

The learning tracks are bounded and verifier-gated, and their datasets are frozen or in a documented, reproducible build state. Missing cells are explicit missing states, never zeros.

4.1 Equivalence

The equivalence track freezes exactly 50,000/5,000/5,000/5,000 base pair records over the four aligned graph channels plus the sequence and floor controls. Source-, e-class-, and trace-relative grouping prevents leakage; IID and OOD views are reported without changing base totals. The pair pipeline has already run end-to-end on a byte-identical 10,000-row pilot corpus, binding real providers (a rule engine and the replay verifier `geml-goal6-rule-engine-replay-verifier-v1`): 9,039 accepted records in 34 CPU minutes — 6,867 replay-verified positives and 2,172 exact-rationally refuted negatives — with 2,824 rejections and 3,133 failures retained under typed reasons, and the shortfall against the frozen production targets reported

explicitly as `complete_short` rather than resampled away (Table 6). One honest quirk is retained rather than patched: `test_ood` yielded only a single negative, because the stress family has few same-family size-matched neighbors.

4.2 Rewrite-step prediction

The rewrite track is one immutable 6-arm $\times$ 3-seed grid: the same goal-conditioned GNN over each of the four channels, one compute-matched prefix transformer, and one uniform sample from the exact registry-derived legal-action mask. We report demonstration-action top- k match *separately* from exact-successor-structure top- k match, together with the verifier-valid proposal rate and every attempted/valid/invalid/unsupported/no-action/timeout/verifier-error denominator. Semantic equivalence is never called demonstration correctness.

4.3 Proof and simplification

The proof track is bounded to exactly 256 frozen proof problems; the simplification track to exactly 1,000 frozen v1 expression IDs. A proof counts as successful only when the retained status is a verified terminal success, the exact target structural signature is reached, the full transition trace replays, the terminal claim is verifier-confirmed, and no invalid transition or verifier error contaminates it; a lower cost, a matching string, or a producer’s claim is not proof success. Uniform, policy, policy-plus-value, and transformer methods share one identical beam/expanded-node/generated-state/proof-depth/wall-time/verifier-call budget, and the primary paired contrast charges every unsuccessful row at least its configured expanded-node budget so that a crash, timeout, or failed proof can never look like an efficiency win. Simplification selects the cheapest visited valid state under one preregistered Goal 3 exact cost (`pure_aml:official_v4`) with a deterministic tie-break, and the goal-conditioned witness-distance value is never used as an undeclared cost.

4.4 Symbolic regression

The SR benchmark is frozen: 256 synthetic tasks plus a 32-task restricted Feynman-style subset (Udrescu and Tegmark, 2020), with a

100-formula exclusion ledger retaining every excluded case and verifier-confirmed recovery as the only recovery criterion. The synthetic tasks were generated and determinism-verified on CPU (byte-identical double builds), and the coordinator freeze recorded canonical manifest checksums for the synthetic-256, development-32, and Feynman-32 sets that reproduced identically across three independent CPU builds.² Production compares an EML-guided arm, an AST-guided arm, a matched PySR or explicitly labeled GP fallback (both implemented; neither ever relabeled as the other), and a compact transformer-SR arm under frozen budgets and three seeds, reporting exact and verifier-confirmed recovery over both attempted and verifier-supported denominators, plus invalidity, timeout, and resource use, and the accuracy–complexity Pareto front rather than best accuracy alone.

5 Domain conformance and external context

5.1 Grammar v2 and the published Gate G10 verdict (C9)

The grammar-v2 workstream adds opt-in `asin/acos/atan/ $\pi$ / $e$` compiler support with the v1 default untouched. Its preregistered bounded conformance audit (74 retained rows, all quotas met, no dropped denominators) was executed on CPU (Apple M1 Pro, Python 3.12.5) at a clean pinned commit and **fails**: zero integrity, coverage, or v1-drift failures, and exactly the eight preregistered `asin/acos` endpoint cells (both compiler modes) nonfinite at 100-digit working precision under the predeclared 10^{-60} tolerances.³ The conformance build was executed three times and produced a byte-identical `records.jsonl` content hash each time.⁴ At 100 digits the 10^{-60} tolerances retain 40 guard digits; independent IEEE end-

²`docs/specs/SR_TASK_SPEC.md` (coordinator freeze 2026-07-30); accepted Feynman count exactly 32.

³`docs/goals/goal10/GOAL10_SUMMARY.md`, “Published verdict”; `FINAL_REPORT.md`
#c9-grammar-v2-conformance.

⁴Content SHA-256 9649faae3991f8c54f8437ac1ab1a9a334606e0499413a048084a72e23da80e9; manifest SHA-256 e25beadd57128a7712bf8edda3d998d5e708cd55930253d37bfb544947ee6ec2; criteria SHA-256 83e305e6f1cb76bbd84f5c877a8d35904f29484fc4cf95c507ce9a404fd91c62.

point probes remain finite and close to the principal references, but because endpoints are required valid source cases, the gate correctly fails rather than hiding them as unsupported or lowering precision until they appear finite. We publish the fail rather than revising criteria or downgrading observed violations to insufficient evidence; no criteria revision is pre-authorized. Gate G10 is conformance-only: it licenses no corpus-v2, expansion, compression, or learning claim.

5.2 Frontier-LLM panel

An external, verifier-normalized reference panel is implemented but is *external context only*: no row it produces may enter the controlled gates G6–G11. The frozen workload is exactly 100 proof and 100 SR tasks per model, 200 attempts per model, using the same two ordered ID lists (hashed and persisted before any call) across four providers, for 800 retained success/failure rows when a model completes. Every provider fact — exact model ID and snapshot, access date, prompts and their hashes, reasoning settings, token and time budgets, supported sampling fields, usage, latency, retries, cost, raw responses, and verifier-confirmed correctness — is recorded. No paid call has been made; the panel ships with paid calls disabled and a zero cost guard, and claimed correctness is kept distinct from verifier-confirmed correctness.

6 Results at fixed scale

Table 2 collects every headline structural number; each row names its committed provenance document. All values come from real CPU production runs recorded in those documents; none is estimated. We organize by outcome, not by a single leaderboard, and give the nulls and the gate fail the space they need to constrain the answer.

6.1 Raw expansion is disqualifying (C2)

Compiling all 250,000 expressions with the pinned official-v4 compiler succeeded on every row and produced a median tree expansion factor $\alpha = 40.6602$ over the source AST, with mean 952.1371, standard deviation 23,155.56, p90 385.07, p99 10,448.60, and max-

imum 6,481,679.31.⁵ The mean/median gap records a heavy right tail — dominated by the trig/hyperbolic family (median 270.46) — not a typical 952-fold expansion; we report the full distribution in Appendix A. Against the six pre-registered per-family thresholds $1 + \ln K / \ln 4L$ (ranging 1.2914–1.5000), and against a whole-grammar counting break-even of ≈ 1.56 , the strict pass count is 0/250,000: no expression in any family is even close to the regime where a reduced operator vocabulary pays for its node count (Table 3). These thresholds are descriptive combinatorial references, not empirical performance laws.

6.2 Exact compression is not competitiveness (C3)

Lossless hash-consed sharing compresses expanded EML trees by a mean factor of 39.3750, yet the shared EML DAG remains larger than the *uncompressed* AST tree for every single one of the 250,000 expressions; the best remaining ratio is exactly 8/7.⁶ The most compressible expression shrinks by 28,087,277/180 and still ends 540/13 times larger than its AST. Mean EML-DAG/AST-tree is 8.3344 and mean EML-DAG/AST-DAG is 10.4750. Per-expression ratios are exact reduced rationals; aggregate means use deterministic compensated `Decimal` accumulation in authoritative corpus order and are labeled approximate. “Compresses well” and “becomes structurally competitive” are different claims, and only the first is true here.

6.3 Verified semantic rewriting helps a little, conditionally (C4)

On a frozen, stratified 30,000-expression selection, bounded equality saturation with complete application provenance and independent candidate verification improves 4,349/18,210 costed rows (23.882%) under branch-insensitive real rewrites (`safe_real`) and 5,026/18,210 (27.600%) with guarded positive-real rules (`positive_real_formal`), at mean relative savings of 2.361% and 2.849% over costed rows and a largest single reduction of 302 exact

⁵docs/goals/goal2/GOAL2_SUMMARY.md, “Result” and “Expansion distribution”; FINAL_REPORT.md #c2-pure-eml-expansion.

⁶docs/goals/goal3/GOAL3_SUMMARY.md, “Structural verdict”; FINAL_REPORT.md #c3-dag-sharing-and-graph-costs.

Quantity	Value	Provenance
Rows processed / exact-count successes (C1)	250,000 / 250,000	Goal 2
Median tree α (EML tree / AST tree) (C2)	40.6602	Goal 2
Mean tree α	952.1371 ($\sigma = 23,155.56$)	Goal 2
p90 / p99 / max α	385.07 / 10,448.60 / 6,481,679.31	Goal 2
Preregistered per-family thresholds 1.2914–1.5000: strict passes	0 / 250,000	Goal 2
Worst family median α (trig/hyperbolic, 40,000 rows)	270.4602	Goal 2
Mean EML self-compression, tree→DAG (C3)	39.3750×	Goal 3
Mean EML DAG / AST tree	8.3344	Goal 3
Mean EML DAG / AST DAG	10.4750	Goal 3
Expressions with EML DAG $\leq$ AST tree	0 / 250,000 (best exactly 8/7)	Goal 3
Largest single self-compression (remaining ratio)	28,087,277/180 (540/13)	Goal 3
E-graph cost coverage (C4)	18,210 / 30,000 (60.700%)	Goal 4
Improved / costed, safe_real	4,349 / 18,210 (23.882%)	Goal 4
Improved / costed, positive_real_formal	5,026 / 18,210 (27.600%)	Goal 4
Mean relative savings over costed rows (incl. unchanged)	2.361% / 2.849%	Goal 4
Unsupported-operator / validation-failure / timeout rows per mode	11,566 / 224 / 0	Goal 4
Null: learned motif vocab. vs. equal-budget frequent (MDL bits, TEST_IID)	324,485,346 vs. 317,678,264	Goal 5
Null: neural ranker vs. structural heuristic (exact-best of 18,050)	8,349 vs. 8,796	Goal 5
Gate G10 conformance verdict (C9)	fail (8 preregistered cells; 74 retained rows)	Goal 10

Table 2: All measured structural results at production scale. Bold marks the winning side or the decisive count. Provenance: committed summaries docs/goals/goal{2,3,4,5,10}/GOAL{2,3,4,5,10}_SUMMARY.md; artifacts are stored off-repository with recorded SHA-256 digests (e.g. Goal 3 run manifest 279b1d01..., analysis fingerprint reproduced twice; Goal 4 run ID 9c26ec30..., rows digest f8fd2e6d...). The committed FINAL_REPORT.md C1–C4/C9 index is a pre-authentication digest index, explicitly not yet final-report-authenticated.

Family	Rows	Median α	Thr.
algebraic core	70,000	25.5778	1.3010
exp/log	40,000	32.7872	1.3382
powers/div/rationals	40,000	44.1579	1.2914
OOD stress	25,000	45.7377	1.3382
mixed elementary	35,000	133.9289	1.4292
trig/hyperbolic	40,000	270.4602	1.5000

Table 3: Per-family median expansion α against the preregistered strict threshold. Every family misses by one to two orders of magnitude; the strict pass count is 0 in each (0/250,000 overall).

EML-DAG nodes.⁷ Coverage is the binding caveat: 11,566 rows per mode (38.553%) contain operators outside the closed rewrite vocabulary — including *all* trig/hyperbolic and mixed-elementary rows — so these results hold

⁷docs/goals/goal4/GOAL4_SUMMARY.md, “Production result”; FINAL_REPORT.md #c4-e-graph-and-motif-compression.

on 60.700% cost coverage and are conditional on it. Each mode also retains 224 validation failures (0.747%), zero timeouts, and zero degraded selections; the source form is always a costed safety anchor, so improvement rates cannot be inflated by dropped rows. The formal library found 677 more improved rows than the safe library, concentrated in **exp_log** and **ood_stress**.

6.4 Two preregistered nulls (C4)

Both Goal 5 comparisons were published in the frozen null-before-positive order.⁸ First, a learned motif vocabulary did *not* beat the equal-budget frequent-motif vocabulary on TEST_IID dictionary-inclusive MDL: 324,485,346 vs. 317,678,264 bits over 25,000 expressions (it did beat the uncompressed

⁸docs/goals/goal5/GOAL5_SUMMARY.md, “Learned and neural comparisons”.

macro baseline, 324,485,346 vs. 572,524,716, and the median of 30 fixed random vocabularies, 324,485,346 vs. 368,943,015). Second, a neural cost ranker did *not* beat every structural heuristic under the frozen comparison order: 8,349/18,050 exact-best rankings vs. 8,796/18,050 for the estimated-EML-tree-cost heuristic (and 8,661/18,050 for the AST-DAG-cost heuristic). Learned components, so far, add nothing over honest structural baselines on these tasks. This is the shape of finding the paper is built to surface: compression that does not translate into competitiveness, and learned machinery that does not beat a transparent rule.

6.5 Learning tracks: verdicts pending

Table 4 states each learning-track verdict. All are `insufficient_evidence` by construction: the apparatus is frozen and CI-tested, but no production training or search has run, and we deliberately do not draft the direction of any result before authenticated evidence exists. The expected cell counts are real (18 equivalence cells, 18 rewrite cells, the 256/1,000 proof/simplification populations, the 256 + 32 SR tasks), and every missing cell is an explicit missing state.

7 Limitations, integrity, and reproducibility

One scale. Everything is measured at one fixed 250k-v1 scale; we fit no scaling law and extrapolate to no other corpus, budget, or grammar. The expressions are synthetic and generated, and the downstream benchmarks are bounded, which limits external validity. With three seeds, planned learning effects will be descriptive, not asymptotically significant.

Bounded semantic audit. Numeric semantic verification covers a deterministic sample, not the whole corpus. The original Goal 2 audit materialized 273 of 250,000 rows (203 passed, 45 nonfinite, 22 overflow, 3 retained mismatches; 7 further selections exceeded the materialization limit).⁹ A widened audit over the byte-identical pilot corpus processed a further 2,988 rows at ≈ 0.085 s/row (2,341 passed,

Track	Result
Equivalence (C5, Goal 6; 6 arms $\times$ 3 seeds)	[RESULTS PENDING — 4 $\times$ H100: equivalence grid: per-arm accuracy vs. AST/sequence/floor]
Rewrite-step (C6, Goal 7)	[RESULTS PENDING — 4 $\times$ H100: demonstration-action and exact-successor top- k ; verifier-valid rate]
Proof & simplification (C7, Goal 8; 256 / 1,000)	[RESULTS PENDING — 4 $\times$ H100: exact-target proof success and exact-cost simplification]
Symbolic regression (C8, Goal 9; 256 + 32)	[RESULTS PENDING — 4 $\times$ H100: EML- vs. AST-guided verifier-confirmed recovery]
Pareto / synthesis (C10–C11, Goal 11)	[RESULTS PENDING — 4 $\times$ H100: fixed-scale quality/compute Pareto; Gate G11 verdict]

Table 4: Learning-track verdicts. Each cell is the paper’s only kind of placeholder: a training number not yet produced. The methodology, budgets, gates, frozen datasets, and denominators for every row are fully specified in §§3–4; only the numbers drop in. Gates G6–G9 and G11 return `insufficient_evidence` until then.

607 nonfinite, 37 overflow, 3 mismatches) before deterministically hanging on the next row — reproduced twice — which is itself a correctness-surface finding: a row whose numeric materialization exceeds practical time, motivating a per-row timeout in the production audit tool. The combined evidence base is thus 3,261 rows with 6 retained mismatches ($\approx 0.1\%$), plus one documented pathological-row hang. We claim structural determinism corpus-wide but numeric verification only for the audited sample; no mismatch is relabeled or resampled away.

Conditional coverage and equal-compute.

The e-graph results hold on 60.7% cost coverage; the excluded families include trig/hyperbolic, the family with the worst expansion, so the semantic-compression numbers cannot be read as corpus-wide. Because pure-EML expansion changes input size and compute, equal parameters are not equal compute in the learning tracks, and we report both sides rather than one.

Conformance, not capability. Gate G10’s published fail bounds the grammar-v2 surface; it neither enables nor blocks any corpus or

⁹docs/goals/goal2/GOAL2_SUMMARY.md, “Semantic audit and retained issues”.

learning result, and remains unresolved by design until a new recorded scope decision. The frontier-LLM panel is external context: proprietary training data, nondeterminism, availability, and model drift keep it out of every controlled gate.

Artifacts and regeneration. Production artifacts are stored off-repository; the repository commits their SHA-256 digests, per-goal manifests, and exact CPU reproduction commands, and every run installs from a single frozen lock file (whose digest was re-frozen to the committed content after a disclosed discrepancy); a CPU-only CI verifies both the floating resolve and the frozen lock environment. Generation reproduces on commodity hardware: the development stage (1,000 rows) regenerated in 42s, and the runner’s own determinism gate ran the 10,000-row pilot twice and verified them byte-identical (corpus hash `df4f3d74...`, zero differences). The 250k stage refuses machines without a twofold memory margin: on a 16 GB laptop it projected a 13,975,961,600-byte peak, required 27.95 GB, found 3.3 GB available, and declined to run degraded — the apparatus enforces its own hardware floor rather than trusting the operator. Certification is incomplete and recorded as such: the binding machinery ran, authenticated zero artifacts in the archive’s absence, and committed a per-claim pre-authentication digest index (27 of 28 manifest entries missing, 35 contractual errors retained) instead of claiming certification it lacks; the license was confirmed canonical MIT under a three-way consistency check. We say so rather than claiming certification.

8 Conclusion

At the fixed 250k scale the compression premise fails quantitatively: median $40.66\times$ expansion against per-family thresholds of at most 1.50, zero AST parity under exact sharing on any of 250,000 expressions, marginal coverage-conditional relief from verified rewriting, and learned components losing both preregistered comparisons. None of this shows the representation is useless for *learning*; it shows that its structural cost is real and measured, so any future learning benefit must pay for it. That question is open, the frozen apparatus to answer it is in place — with its pair pipeline

already exercised on a pilot corpus — and the published record, two nulls and one gate fail, none softened, is the standard any answer will be held to.

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A Full expansion distribution

Table 5 gives the complete percentile profile of the tree expansion factor α over all 250,000 rows, from the Goal 2 production run (clean commit `4a5e8b9e...`, eight workers, 2,557.26s wall). The single largest case has 13 AST nodes and 84,261,831 exact EML nodes ($\alpha = 6,481,679.31$).

Statistic	Tree α
minimum	1.5000
p10	21.0338
p25	26.3800
median	40.6602
p75	75.2550
p90	385.0723
p95	1,187.8240
p99	10,448.5978
maximum	6,481,679.3077
mean	952.1371
std. dev.	23,155.5594

Table 5: Expansion factor $\alpha = (\text{exact EML tree nodes})/(\text{source AST nodes})$ over all 250,000 rows (Goal 2).

B Pilot pair build

Table 6 gives the per-split counts from the real 34-CPU-minute pilot pair build over the byte-identical pilot corpus (config `sha256:377d983b...`, status `complete_short`). Every positive carries a replayable verifier trace and every negative is formally refuted at exact-rational probe points. Shortfalls are against the frozen 50,000/5,000/5,000/5,000 production targets and are reported, not resampled away; materializing one graph channel over these pairs yields 18,078 per-channel tensors (two endpoints per pair).

Split	Acc. (pos+neg)	Rej.	Fail	Short
train	6,281 (4,568+1,713)	1,786	2,432	43,719
validation	893 (657+236)	263	343	4,107
test_iid	864 (642+222)	276	358	4,136
test_ood	1,001 (1,000+1)	499	0	3,999
total	9,039 (6,867+2,172)	2,824	3,133	—

Table 6: Pilot pair build (Goal 6 pipeline). Accepted = positives + negatives; rejected and failed rows are retained with typed reasons. The lone `test_ood` negative reflects few same-family size-matched neighbors in the stress family and is reported, not patched.